HEALTH-AI ML Learning Tool

Sprint 2 – QA Final Test Report

Test Environment

Application: HEALTH-AI ML Learning Tool

Test Type: Manual UI Testing

Environment: Localhost development environment

Browser: Google Chrome

Operating System: Windows

Tester: Berfin Duru Alkan

Course: Software Quality Assurance

Date:15.03.2026

Test Execution Summary

Status	Count
PASS	28
TOTAL	28

STEP 1 — Clinical Context

TC-S2-001 — Step 1 default page loads successfully

Priority: High

Precondition: App opened on initial screen

Steps:

1. Open the application
2. Observe Step 1 content
3. Verify title, context card and stepper

Expected Result:
Step 1 opens by default with Clinical Context visible and domain pills displayed.

Current Status: PASS

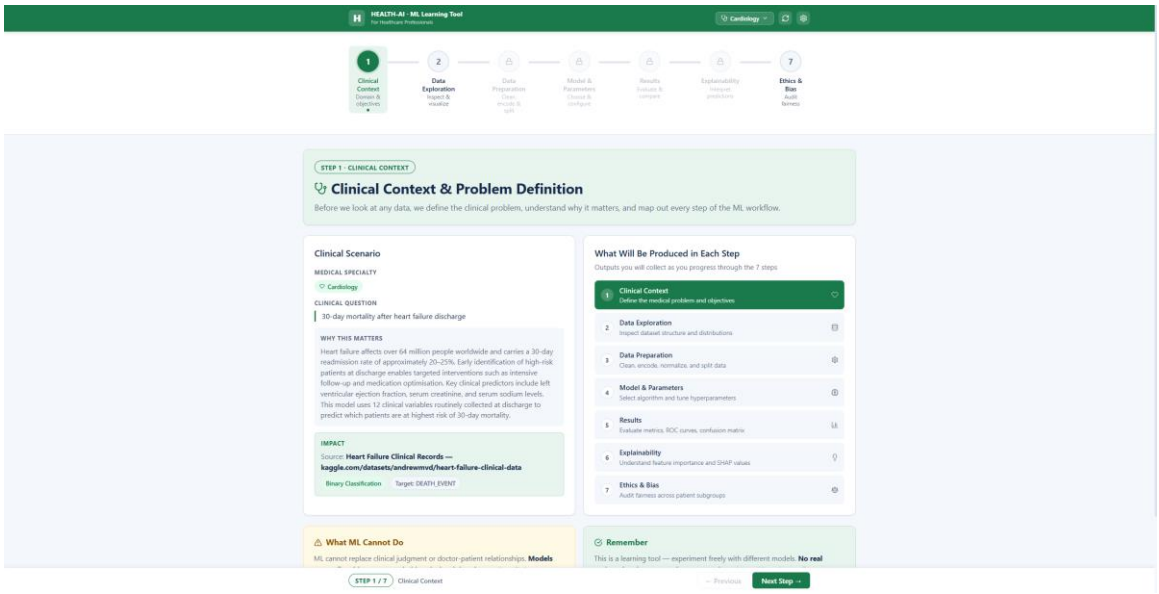

Figure 1

TC-S2-02 — Progress bar correctly highlights current step

Priority: Medium

Precondition: User is on any step of the pipeline

Steps:

1. Open the application
2. Navigate to different steps of the tool
3. Observe the progress bar / stepper

Expected Result:

The current step is clearly highlighted

Steps 1, 2, and 7 remain accessible regardless of pipeline state

The progress indicator updates correctly when navigating between steps

Current Status: PASS

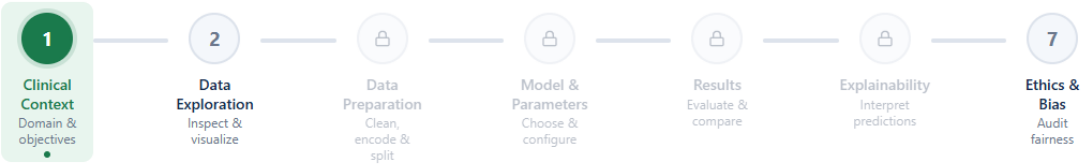

Figure 2

TC-S2-003 — Domain pill click updates Step 1 clinical text

Priority: Critical

Precondition: App on Step 1

Steps:

1. Click different domain pill
2. Observe Step 1 title/content update

Expected Result:
Selected domain becomes active and clinical scenario text updates.

Current Status: PASS

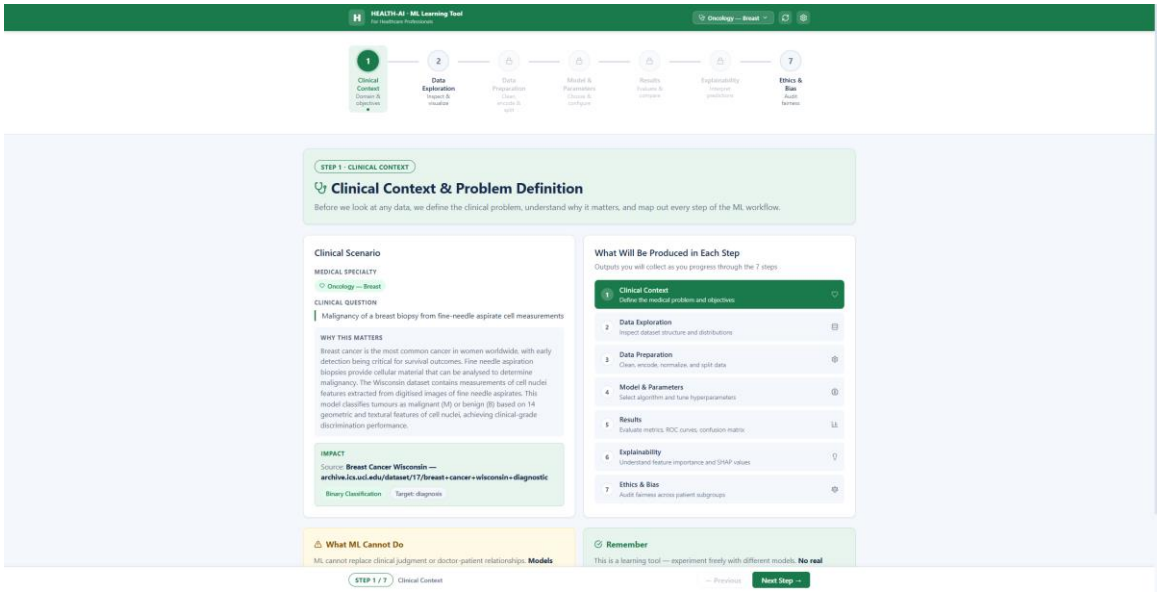

Figure 3

TC-S2-004 — Domain switch confirmation appears after progress exists

Priority: Critical

Precondition: User progressed in pipeline

Steps:

1. Enter Step 2 or Step 3
2. Click different domain pill

Expected Result:
Confirmation dialog warns about pipeline reset.

Current Status: PASS

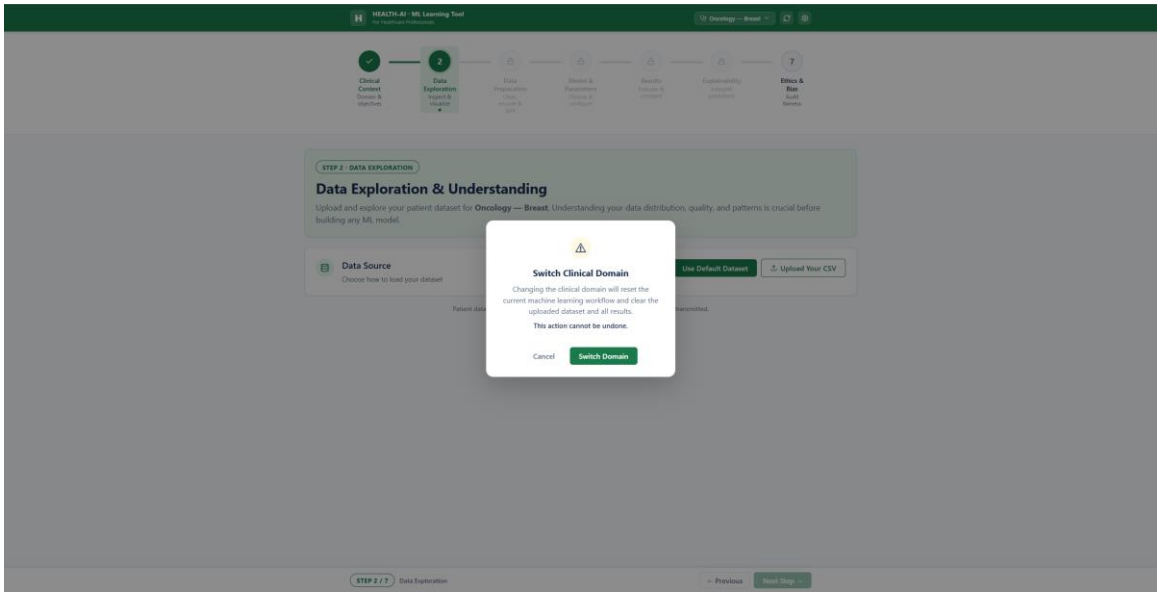

Figure 4

TC-S2-005 — All 20 domains are present in domain selection

Priority: Critical

Precondition: App open

Steps:

1. Count domain pills
2. Compare with User Guide list

Expected Result:
20 clinical domains should be visible.

Current Status: PASS

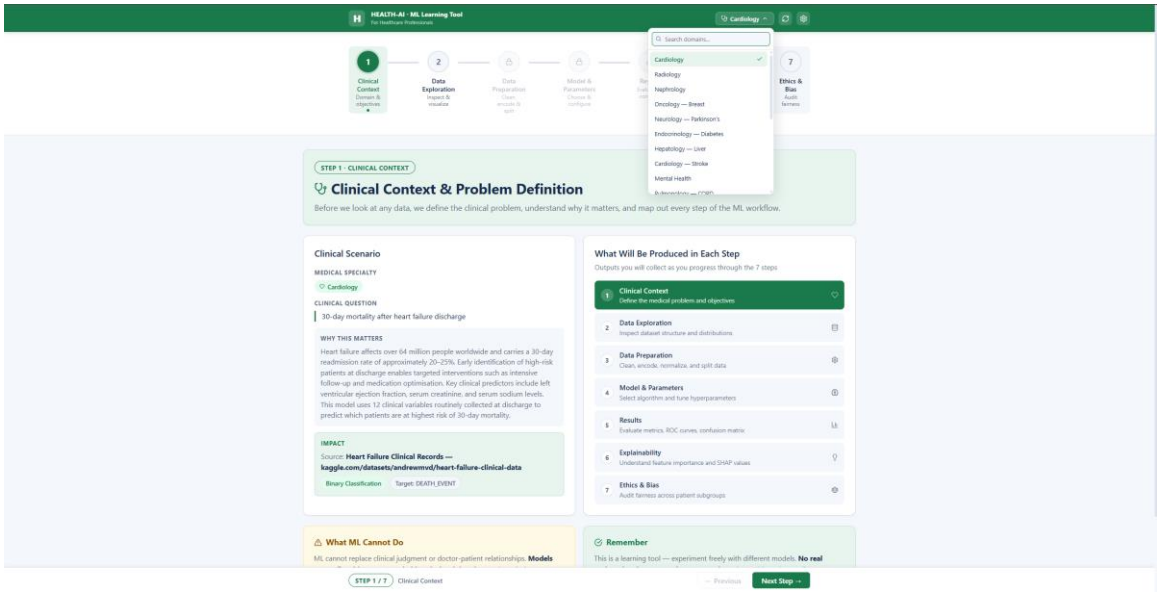

Figure 5

STEP 2 — Data Exploration

TC-S2-006 — Built-in example dataset loads successfully

Priority: Critical

Precondition: User on Step 2

Steps:

1. Click Use Default Dataset

Expected Result:
Dataset loads and charts + summaries appear.

Current Status: PASS

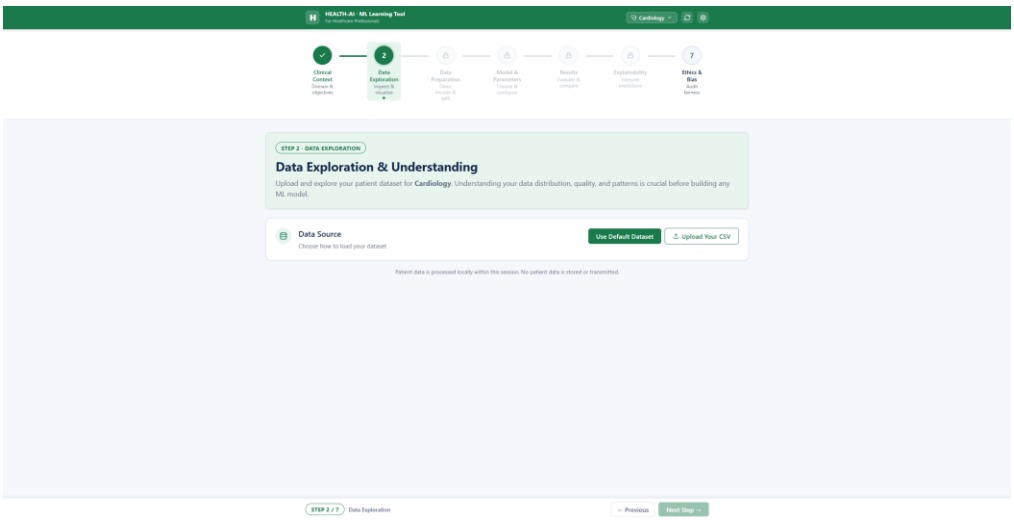

Figure 6.1

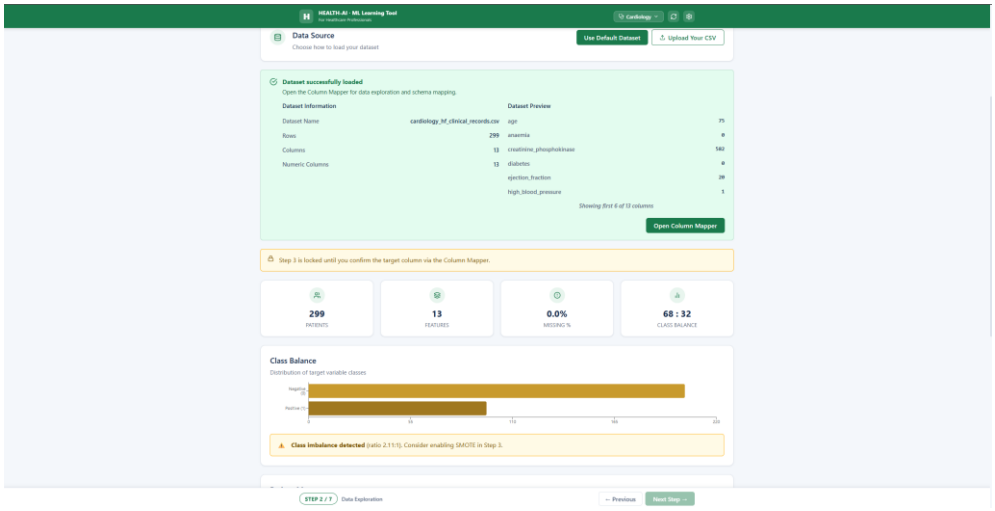

Figure 6.2

TC-S2-007 — Step 2 right-panel content is displayed

Priority: Critical

Precondition: Example dataset loaded

Steps:

1. Check right panel

Expected Result:
Class distribution, dataset summary and column summary visible.

Current Status: PASS

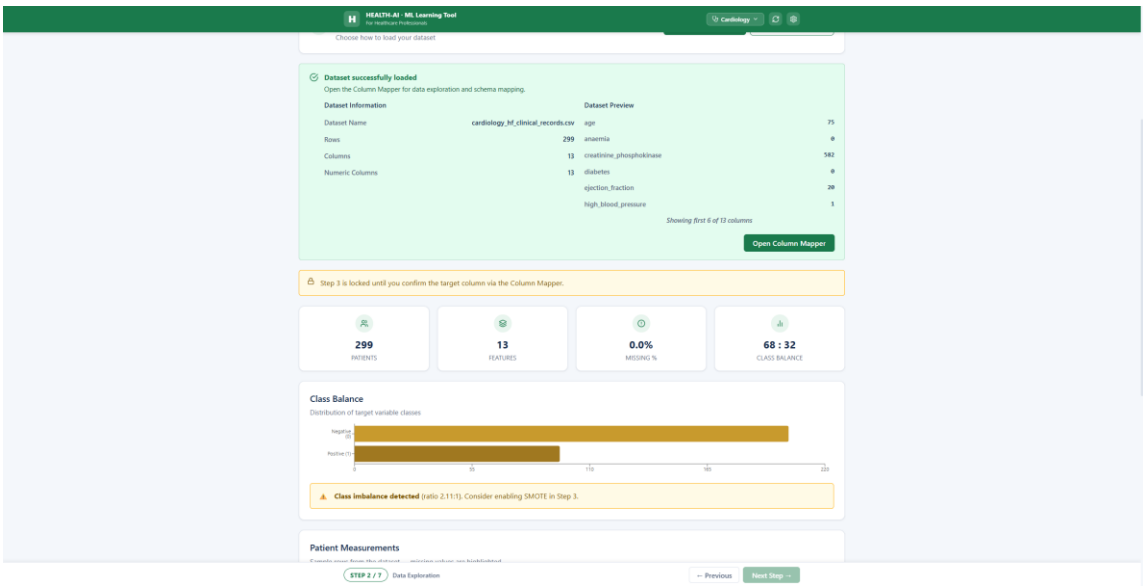

Figure 7

TC-S2-008 — Privacy notice visible

Priority: Medium

Precondition: User on Step 2

Steps:

1. Observe dataset area

Expected Result:
Privacy banner visible explaining browser-only processing.

Current Status: PASS

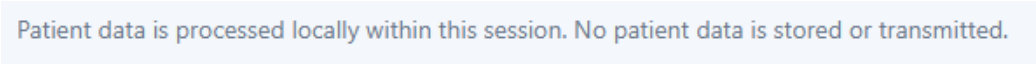

Figure 8

TC-S2-009 — Valid CSV upload accepted

Priority: Critical

Precondition: User on Step 2

Steps:

1. Upload valid CSV (<50MB, numeric column)

Expected Result:
CSV accepted and parsed.

Current Status: PASS

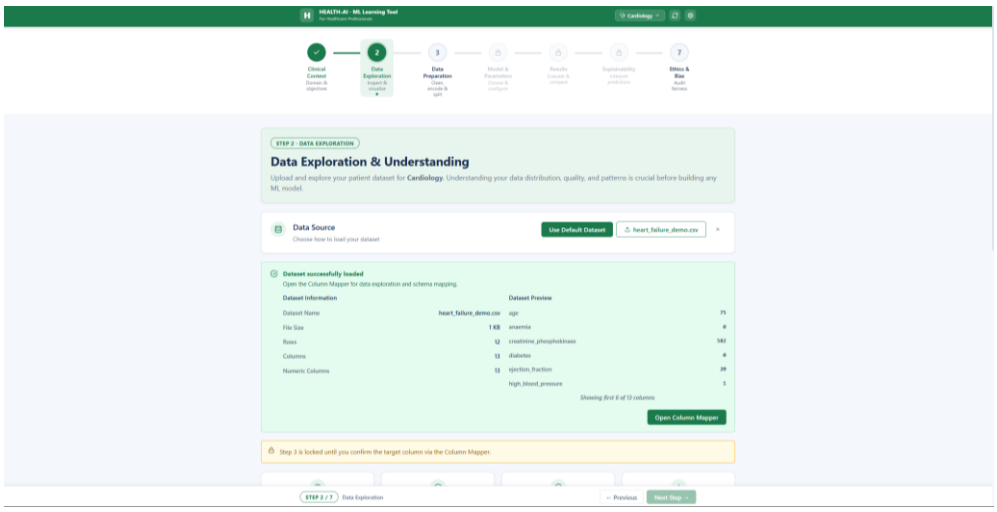

Figure 9

TC-S2-010 — Invalid file type rejected

Priority: Critical

Precondition: User on Step 2

Steps:

1. Upload .txt/.xlsx file

Expected Result:
Upload rejected with friendly error.

Current Status: PASS

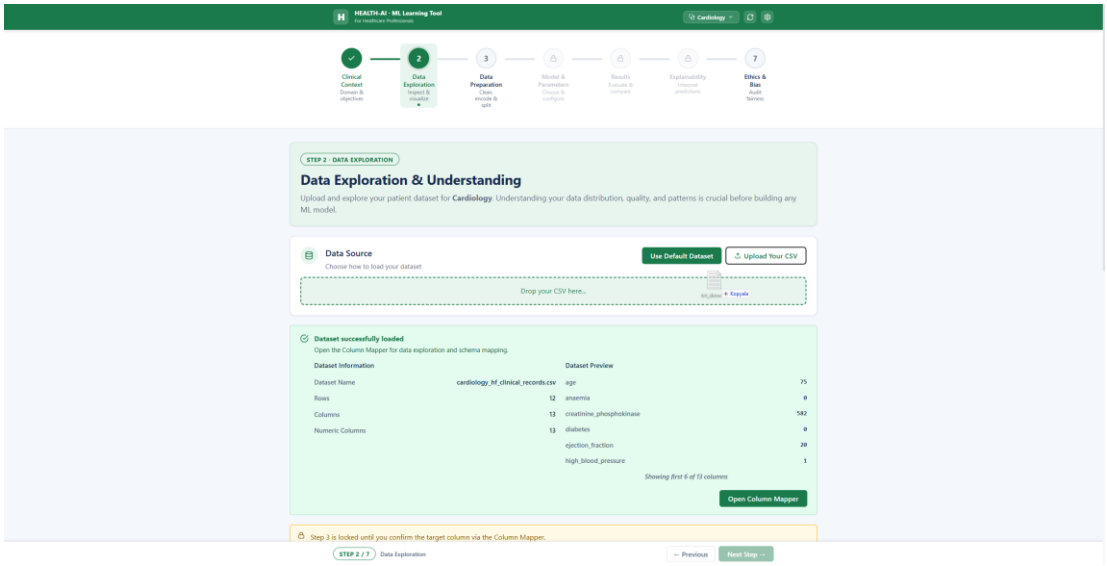

Figure 10.1

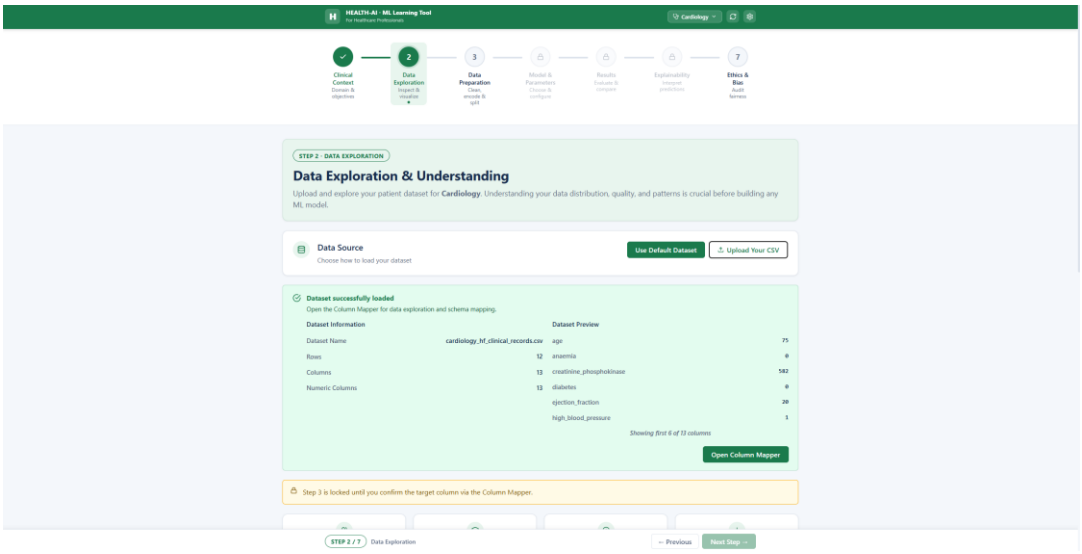

Figure 10.2

TC-S2-011 — CSV with fewer than 10 rows rejected

Priority: High

Precondition: User on Step 2

Steps:

1. Upload small CSV (1-9 rows)

Expected Result:
Upload rejected with minimum row message.

Current Status: PASS

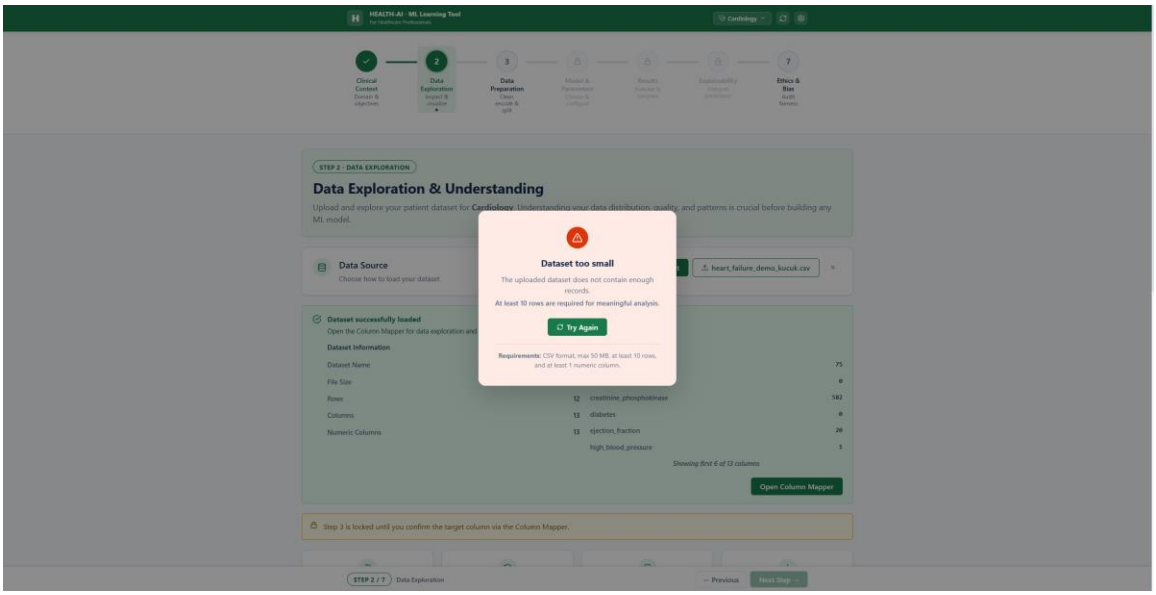

Figure 11

TC-S2-012 — CSV without numeric column rejected

Priority: High

Precondition: User on Step 2

Steps:

1. Upload CSV without numeric fields

Expected Result:
Upload rejected with numeric requirement message.

Current Status: PASSs

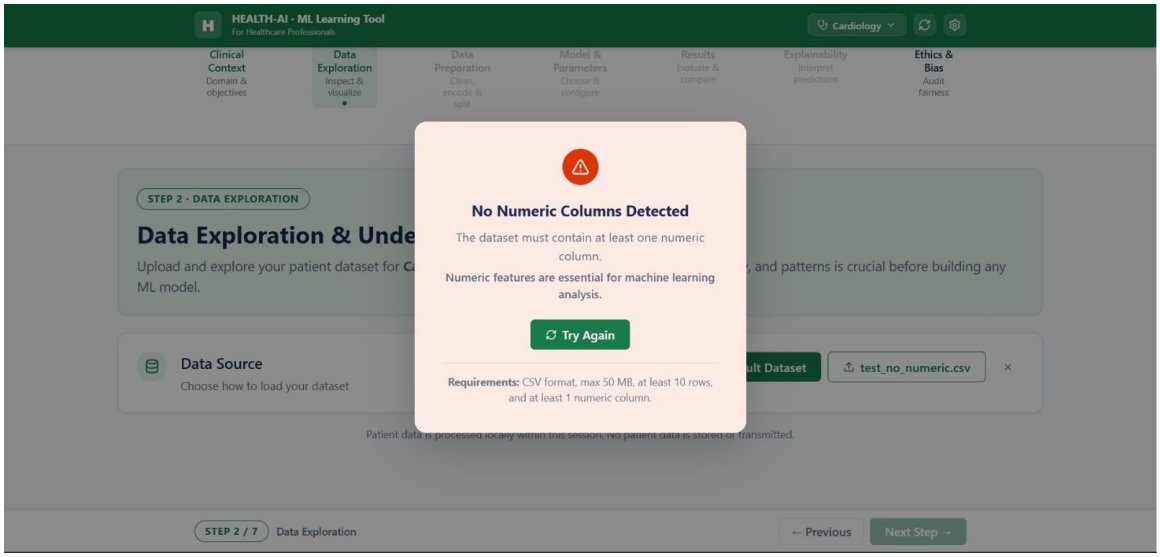

Figure 12

TC-S2-013 — CSV without numeric column rejected

Priority: High

Precondition: User on Step 2

Steps:

1. Navigate to Step 2 — Data Exploration
2. Do not open or complete the Column Mapper
3. Click Step 3 — Data Preparation from the stepper

Expected Result:
Step 3 should remain locked until the user validates and saves the target column in Column Mapper.

Current Status: PASS

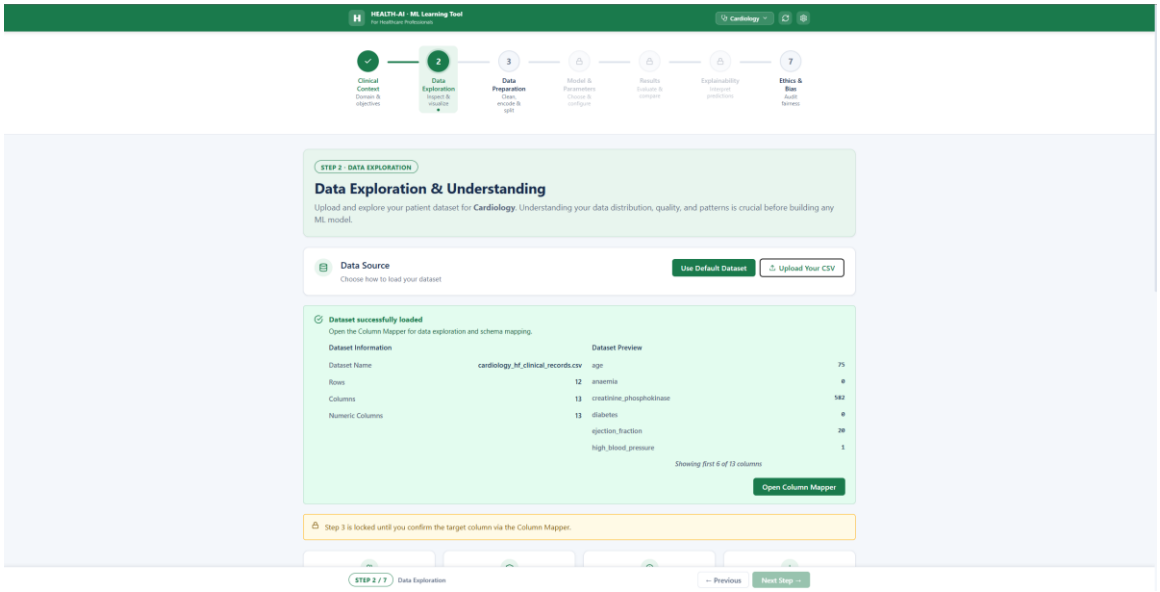

Figure 13

TC-S2-014 — Feature table shows missing/type/action info

Priority: High

Precondition: Dataset loaded

Steps:

1. Inspect column summary

Expected Result:
Column name/type/missing/unique/sample values visible.

Current Status: PASS

Patient Measurements

Sample rows from the dataset — missing values are highlighted

COLUMN NAME	TYPE	MISSING	UNIQUE VALUES	SAMPLE VALUES
age	int64	0 (0%)	10	75, 55, 65, 50
anaemia	int64	0 (0%)	2	0, 0, 0, 1
creatinine_phosphokinase	float64	Missing (16.67%)	10	582, 7861, 146, 111
diabetes	float64	Missing (25%)	2	0, 0, 0, 1
ejection_fraction	float64	Missing (8.33%)	6	20, 38, 20, 20
high_blood_pressure	int64	0 (0%)	2	1, 0, 0, 0
platelets	float64	Missing (16.67%)	10	265000, 263358.03, 210000, 327000
serum_creatinine	float64	Missing (25%)	6	1.9, 1.1, 1.9, 2.1
serum_sodium	float64	Missing (8.33%)	11	130, 136, 129, 137
sex	float64	Missing (8.33%)	2	1, 1, 1, 0
smoking	int64	0 (0%)	2	0, 0, 1, 0
time	int64	0 (0%)	10	4, 6, 7, 7
DEATH_EVENT target	int64	0 (0%)	2	1, 1, 1, 0

Figure 14

COLUMN MAPPER / GATE LOGIC

TC-S2-015 — Step 3 blocked before Column Mapper save

Priority: Critical

Precondition: Dataset loaded but mapping not saved

Steps:

1. Attempt navigation to Step 3

Expected Result:
Step 3 locked with warning banner.

Current Status: PASS

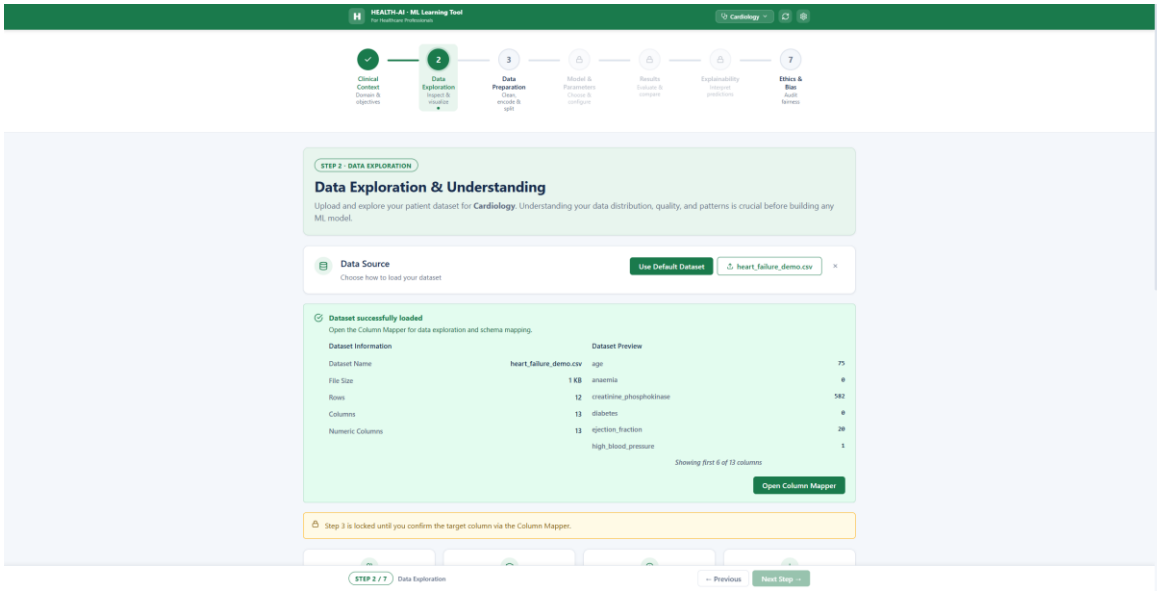

Figure 15

TC-S2-016 — Column Mapper opens from Step 2

Priority: Critical

Precondition: Dataset loaded

Steps:

1. Click Open Column Mapper

Expected Result:
Modal opens allowing target column selection.

Current Status: PASS

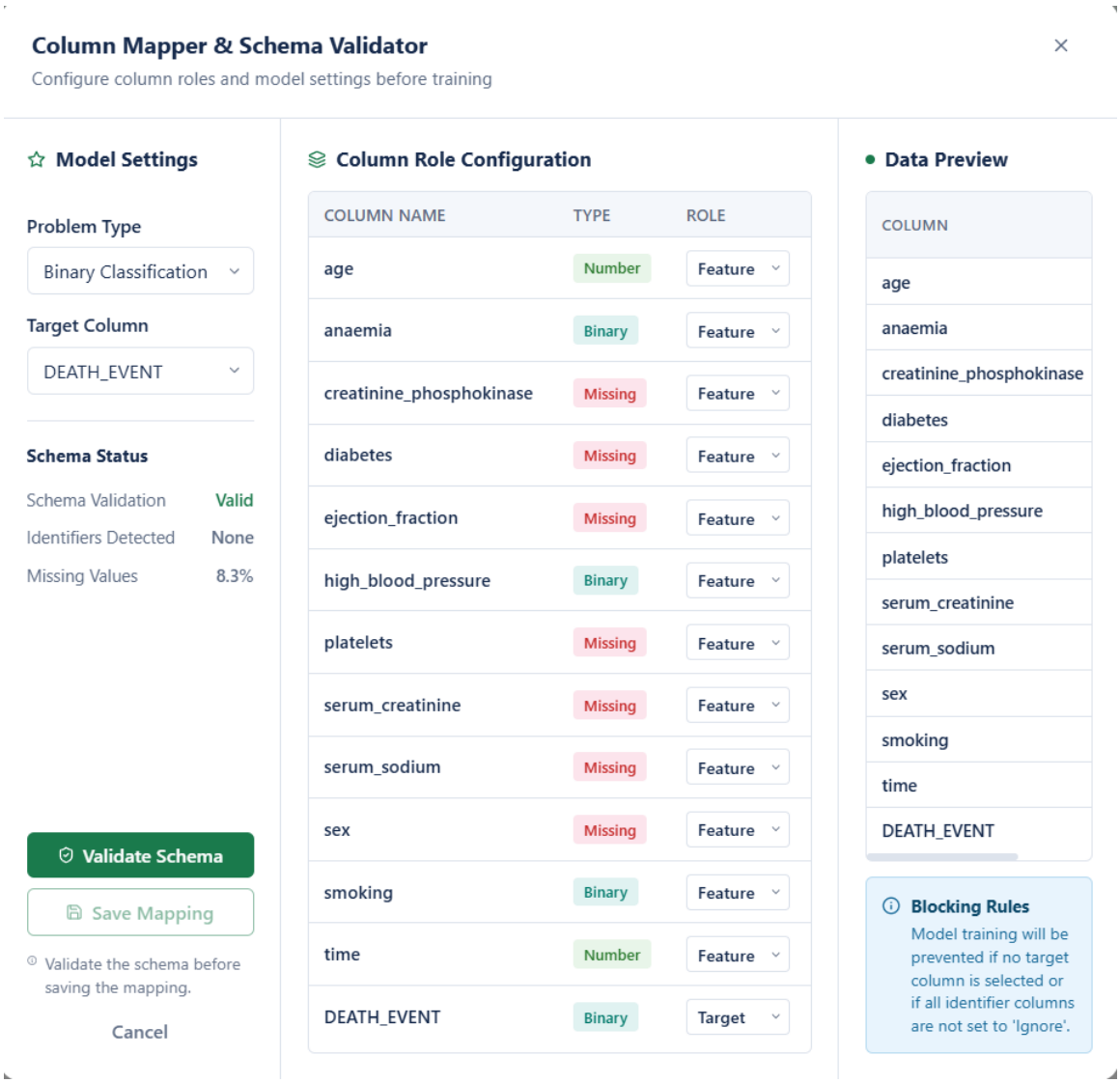

Figure 16

TC-S2-017 — Validate then Save mapping unlocks Step 3

Priority: Critical

Precondition: Column Mapper open

Steps:

1. Select target column
2. Validate
3. Save

Expected Result:
Mapping saved and Step 3 unlocked.

Current Status: PASS

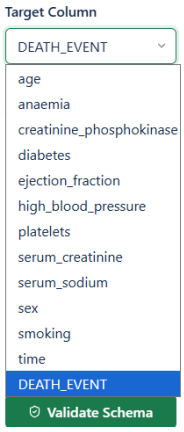

Figure 17

TC-S2-018 — Save success banner shown

Priority: High

Precondition: Mapping saved

Steps:

1. Save mapping

Expected Result:
Green confirmation banner appears.

Current Status: PASS

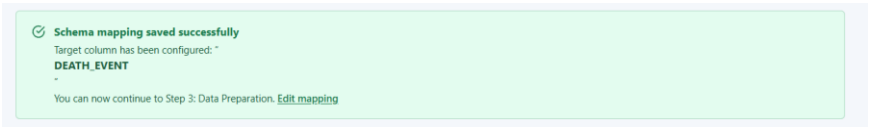

Figure 18

TC-S2-019 — Bypass attempt without save blocked

Priority: Critical

Precondition: Mapper cancelled

Steps:

1. Cancel mapper
2. Try Step 3

Expected Result:
Step 3 remains locked.

Current Status: PASS

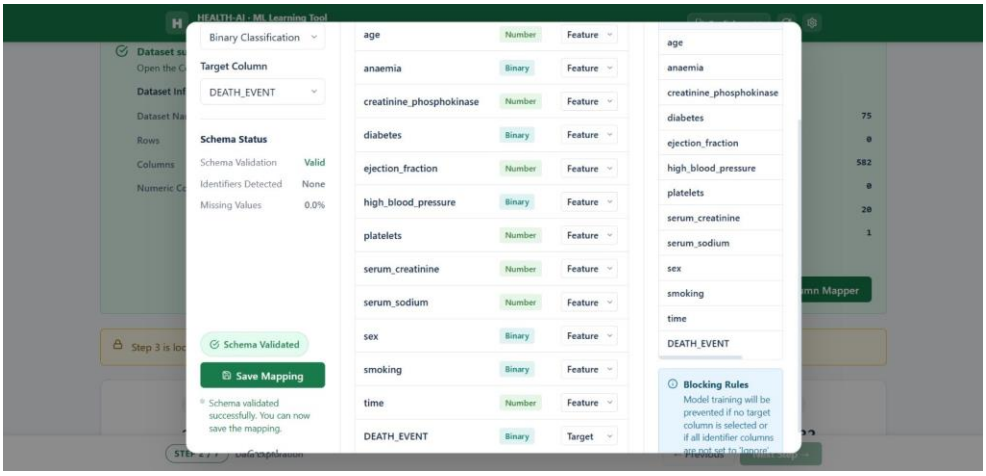

Figure 19.1

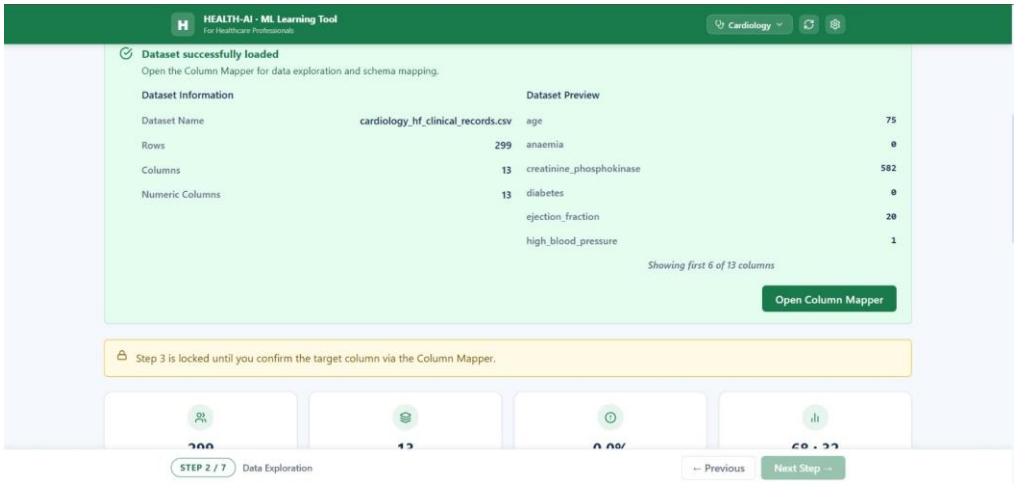

Figure 19.2

STEP 3 — Data Preparation

TC-S2-020 — Step 3 page opens after Step 2 validation

Priority: Critical

Precondition: Mapping saved

Steps:

1. Navigate to Step 3

Expected Result:
Preparation controls visible.

Current Status: PASS

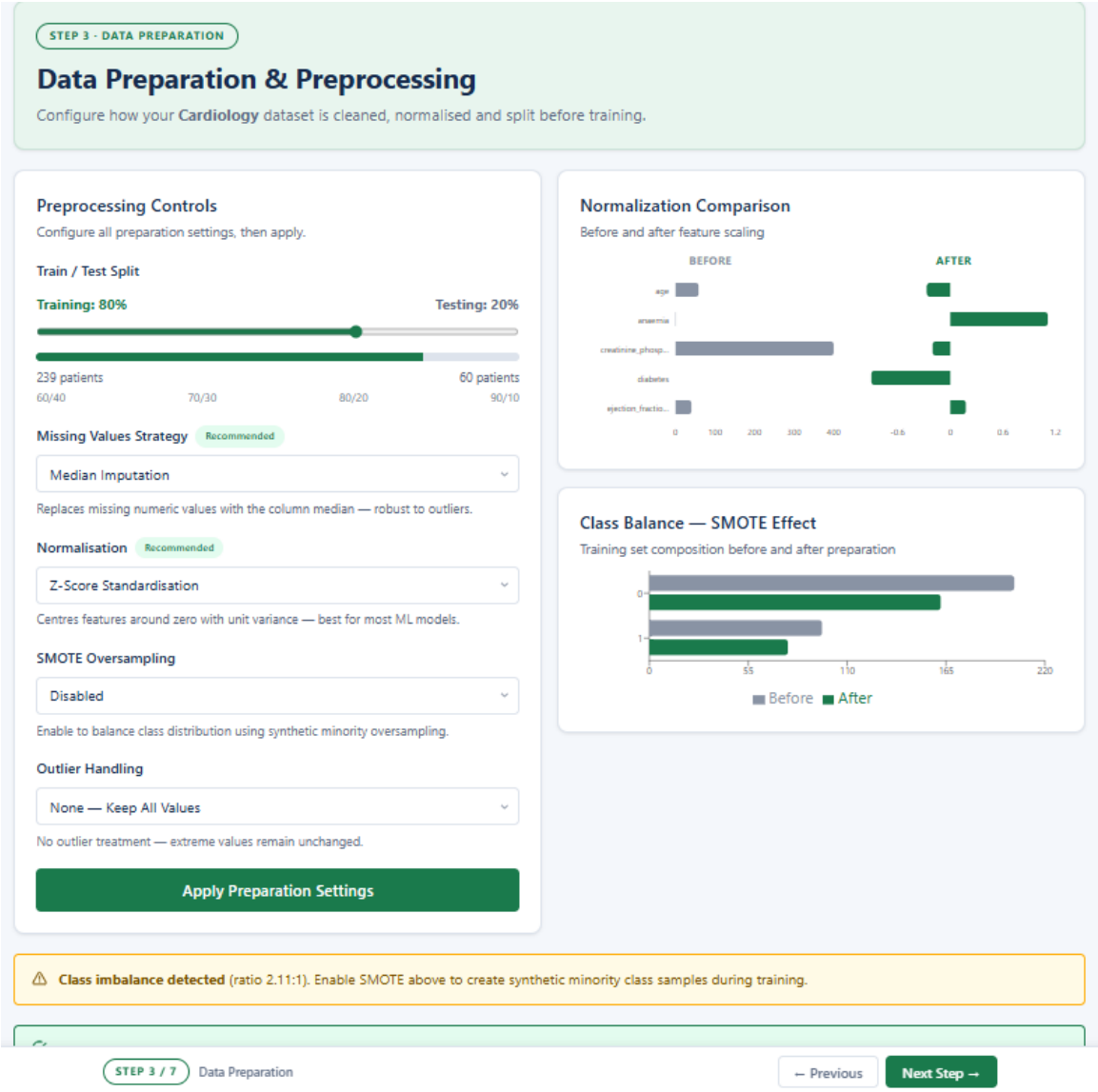

Figure 20

TC-S2-021 — Train/Test slider functional

Priority: Critical

Precondition: User on Step 3

Steps:

1. Move slider

Expected Result:
Percentages update correctly.

Current Status: PASS

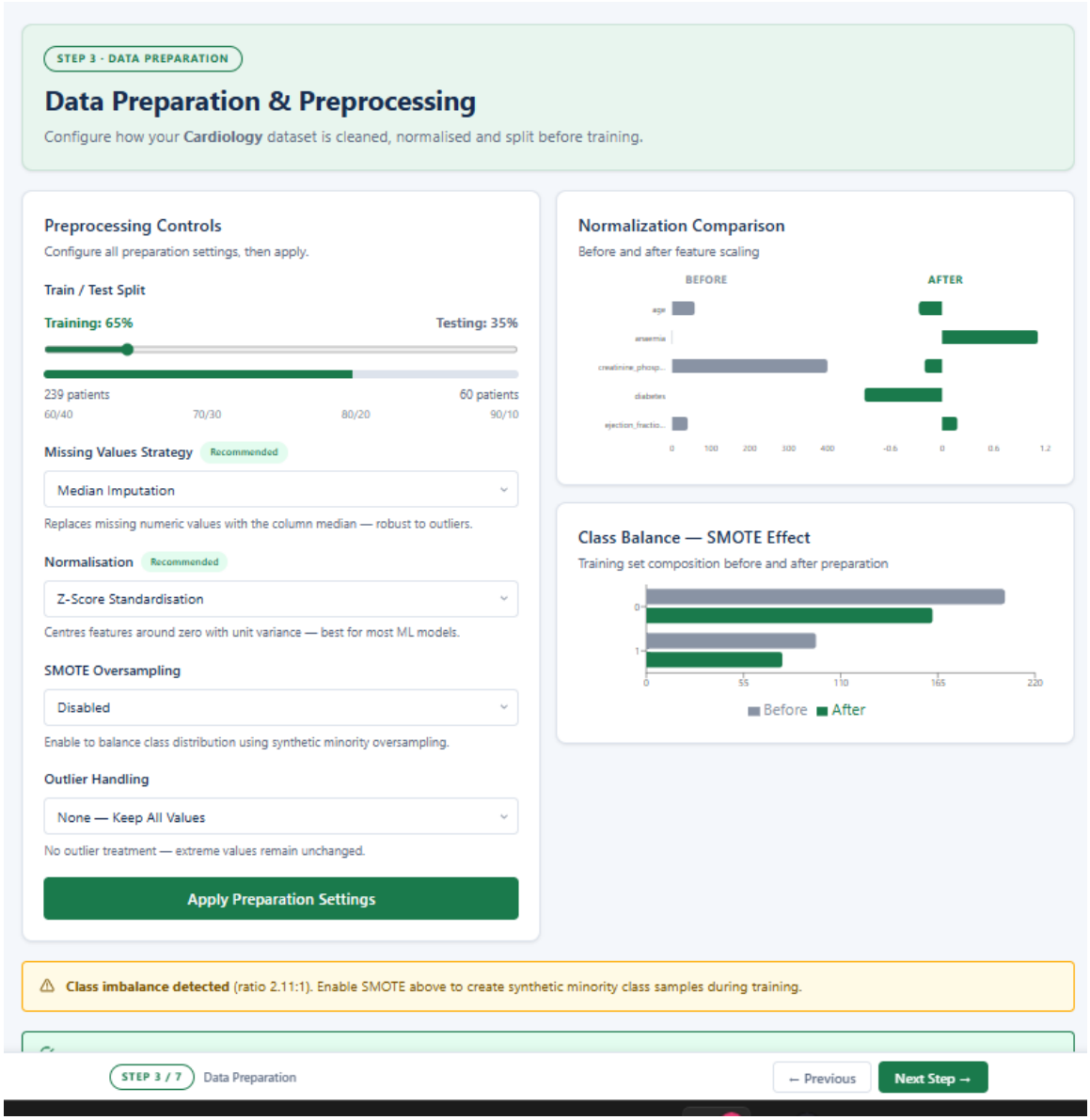

Figure 21

TC-S2-022 — Missing values control functional

Priority: Critical

Precondition: User on Step 3

Steps:

1. Change missing value option

Expected Result:
Selection changes correctly.

Current Status: PASS

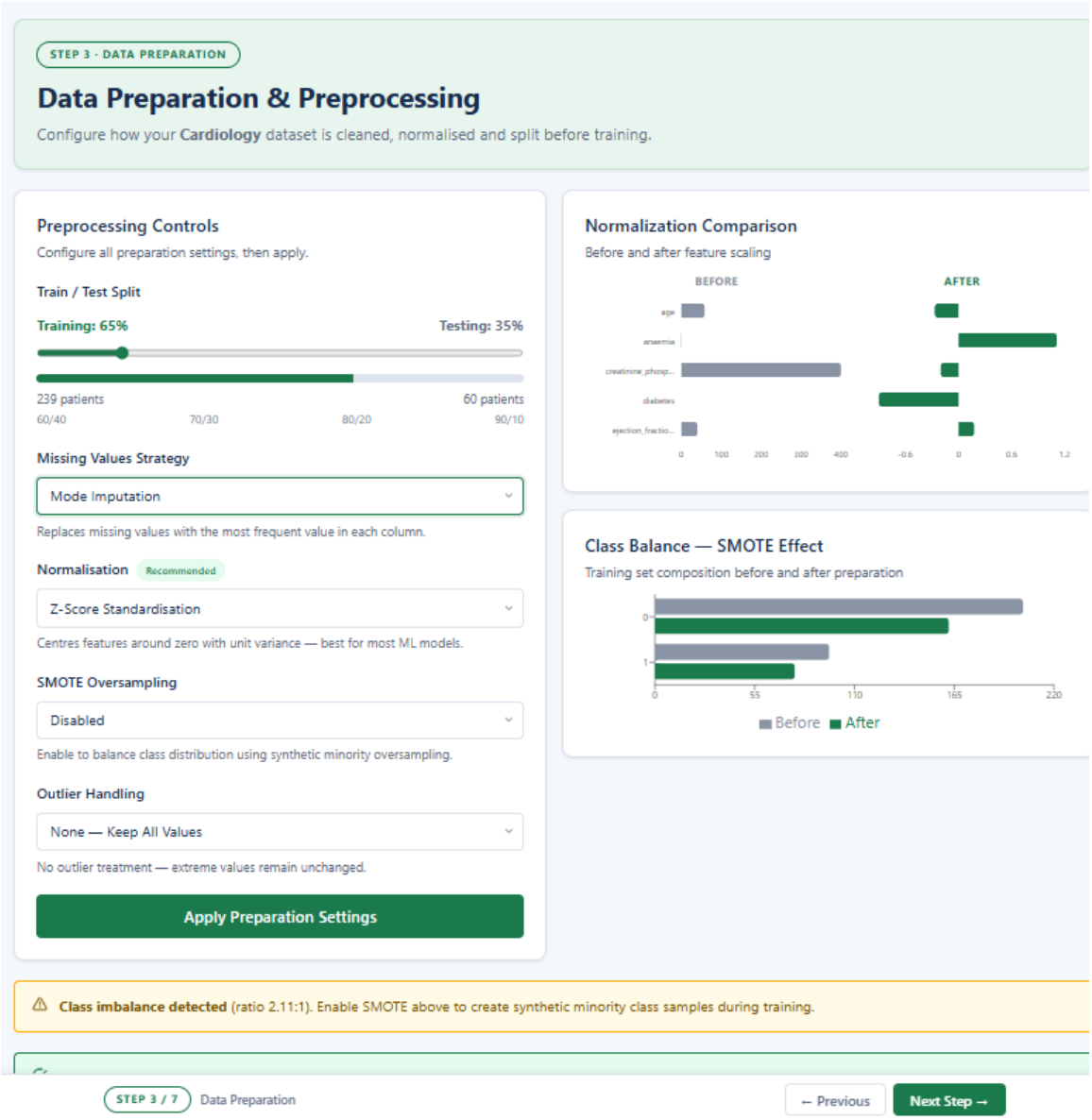

Figure 22

TC-S2-023 — Normalization control functional

Priority: Critical

Precondition: User on Step 3

Steps:

1. Switch Z-score / Min-Max / None

Expected Result:
Selection changes correctly.

Current Status: PASS

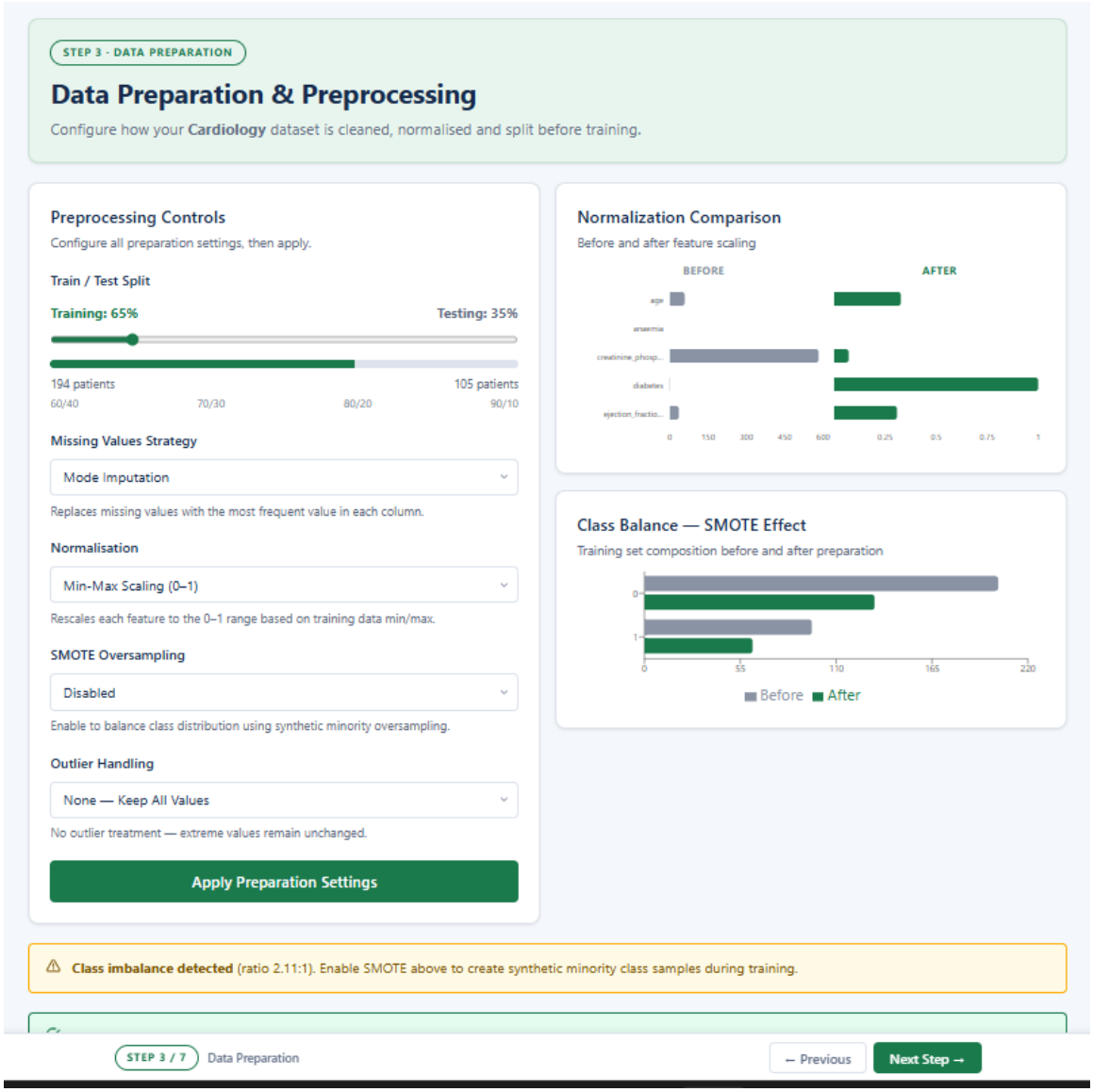

Figure 23

TC-S2-024 — Class imbalance control functional

Priority: Critical

Precondition: Dataset has imbalance

Steps:

1. Open Step 3 imbalance control

Expected Result:
Options selectable.

Current Status: PASS

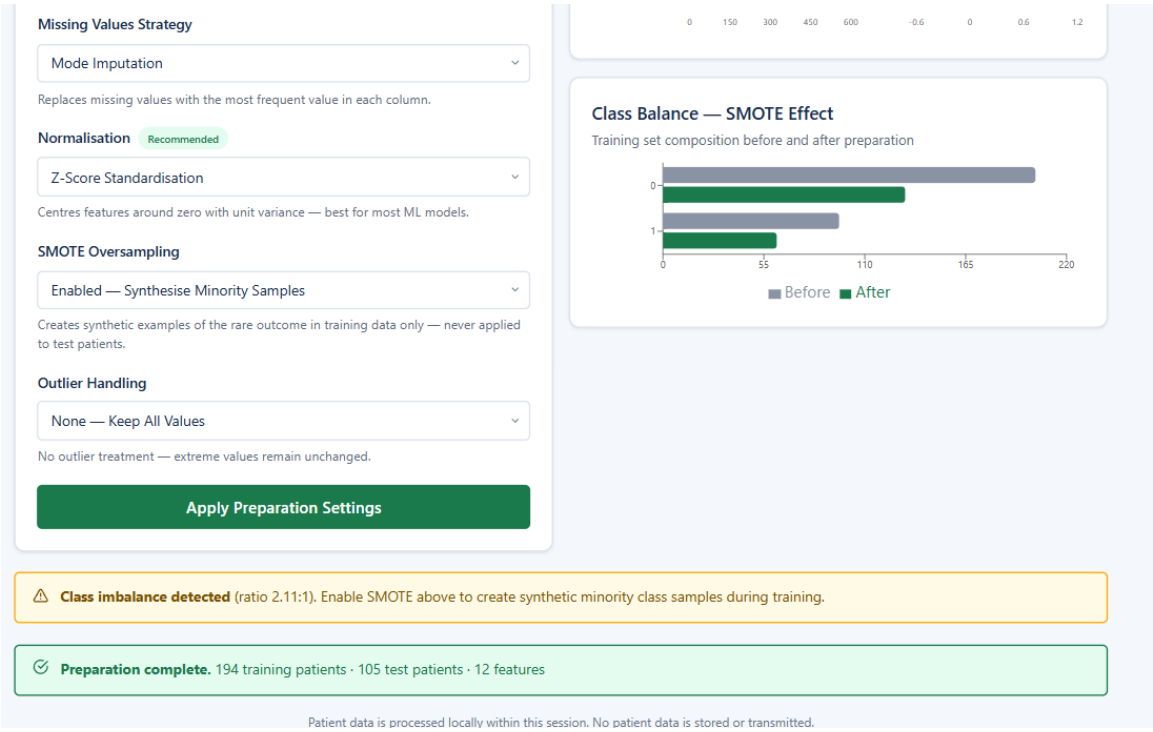

Figure 24

TC-S2-025— Apply Preparation Settings button works

Priority: Critical

Precondition: Options selected

Steps:

1. Click Apply Preparation Settings

Expected Result:
Preparation executes successfully.

Current Status: PASS

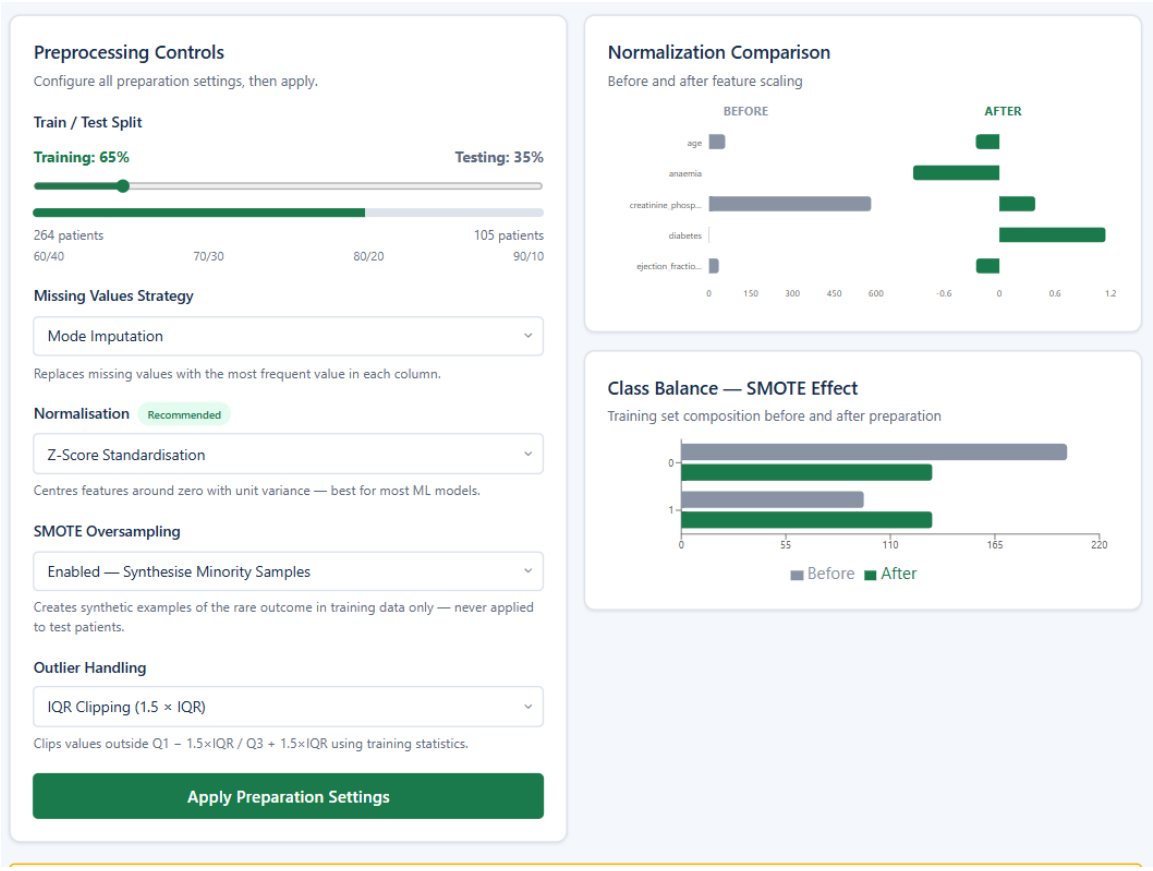

Figure 25

TC-S2-026 — Before/after normalization chart shown

Priority: Critical

Precondition: Preparation applied

Steps:

1. Observe charts

Expected Result:
Before/after normalization chart visible.

Current Status: PASS

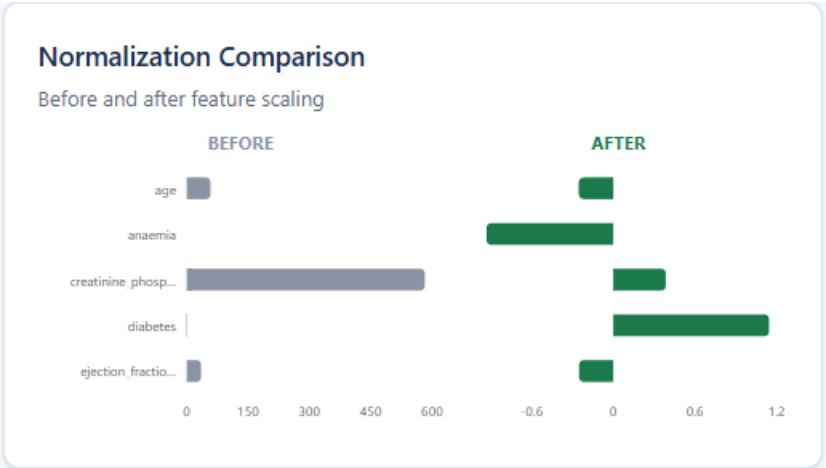

Figure 26

TC-S2-027 — Before/after SMOTE class balance chart shown

Priority: Critical

Precondition: Imbalanced dataset

Steps:

1. Apply preprocessing

Expected Result:
Before/after SMOTE chart visible.

Current Status: PASS

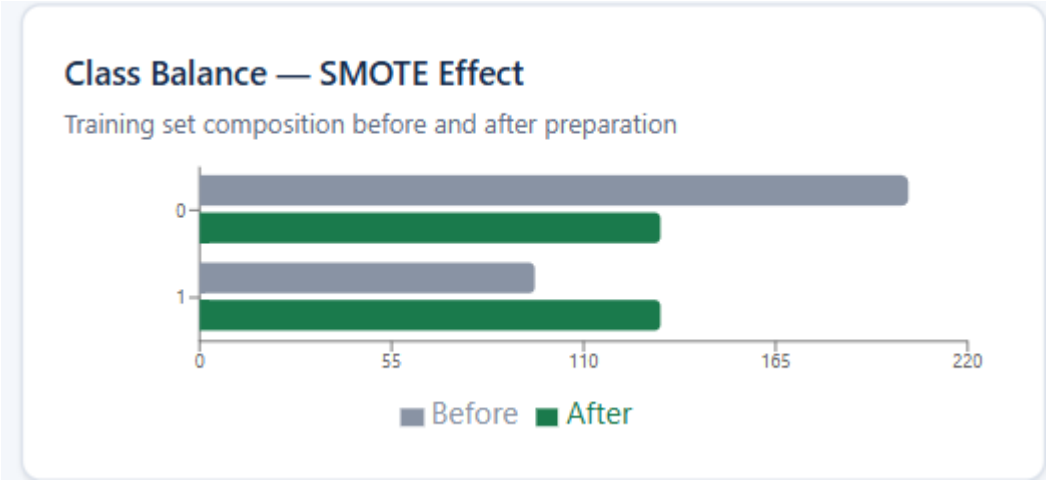

Figure 27

TC-S2-028 — Green success banner after apply

Priority: High

Precondition: Preparation applied

Steps:

1. Click Apply

Expected Result:
Green success banner confirming preprocessing.

Current Status: PASS

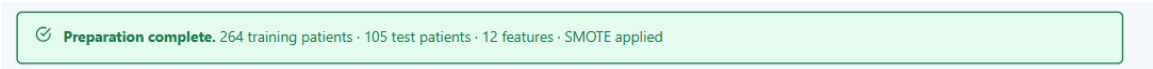

Figure 28